

Problem 1: Put 6 counters in piles of 2, 2, 1 and 1. A move goes like this: take one counter from every pile, and put those counters together as one new pile. Make moves, and after each move write the pile sizes in the next row, biggest first. Stop when a row is exactly the same as the row above it. How many moves did it take until the piles stopped changing?

Move	Piles, biggest first
Start	2 2 1 1
1	
2	
3	
4	
5	
6	
7	
8	

Moves until it stopped: _____

Problem 2: Do the same from three more starts: two piles of 3; one pile of 6; six piles of 1. Write the number of moves for each start.

Move	Piles, biggest first
Start	3 3
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 3 3: _____

Move	Piles, biggest first
Start	6
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 6: _____

Move	Piles, biggest first
Start	1 1 1 1 1 1
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 1 1 1 1 1 1: _____

Problem 3: There are exactly 11 ways to arrange 6 counters in piles. They are all listed here. Next to each one, write how many moves it takes until the piles stop changing. You already know four of them; run the rest with your counters.

6	_____	3 1 1 1	_____
5 1	_____	2 2 2	_____
4 2	_____	2 2 1 1	_____
4 1 1	_____	2 1 1 1 1	_____
3 3	_____	1 1 1 1 1 1	_____
3 2 1	_____		

Problem 4: Now 8 counters. Put all 8 in one pile. Make moves and write each row. Stop when a row is the same as an earlier row, not just the row above it. Draw a bracket beside the rows that form the loop. How many arrangements are in the loop?

Move	Piles, biggest first
Start	8
1	
2	
3	
4	
5	
6	
7	
8	
9	

Arrangements in the loop: _____

Problem 5: Start 8 counters as four piles of 2, and then as piles of 5, 1, 1, 1. For each start, make moves and write rows until a row repeats an earlier row. Which of your three 8-counter games go round on the same arrangements?

Move	Piles, biggest first
Start	2 2 2 2
1	
2	
3	
4	
5	
6	
7	

Move	Piles, biggest first
Start	5 1 1 1
1	
2	
3	
4	
5	
6	
7	

Same loop as the game from one pile of 8: _____

Problem 6: Here are all 22 arrangements of 8 counters. Share them out at your table. Run each one on a blank sheet until a row is an arrangement you have already seen in a loop or in a finished run. Write L next to it if it reaches the loop of 4 3 1, S if it reaches the loop of 4 2 2, and N if it goes round on arrangements in neither loop. Count how many arrangements end in each loop.

8	_____	3 3 2	_____
7 1	_____	3 3 1 1	_____
6 2	_____	3 2 2 1	_____
6 1 1	_____	3 2 1 1 1	_____
5 3	_____	3 1 1 1 1 1	_____
5 2 1	_____	2 2 2 2	_____
5 1 1 1	_____	2 2 2 1 1	_____
4 4	_____	2 2 1 1 1 1	_____
4 3 1	_____	2 1 1 1 1 1 1	_____
4 2 2	_____	1 1 1 1 1 1 1 1	_____
4 2 1 1	_____		
4 1 1 1 1	_____		

Long loop: _____ arrangements

Short loop: _____ arrangements

Did any start end in a new loop? _____

Problem 7: Run 4 counters from one pile of 4, and 5 counters from one pile of 5, until a row repeats an earlier row. Then list all seven arrangements of 5 counters in the space on the right. Use your list to explain why every game with 5 counters must repeat an arrangement by move 7 at the latest.

Move	Piles, biggest first
Start	4
1	
2	
3	
4	
5	

Move	Piles, biggest first
Start	5
1	
2	
3	
4	
5	
6	

Problem 8: Run 10 counters from three starts: one pile of 10; two piles of 5; five piles of 2. Stop each run when a row is the same as the row above it, and write the number of moves it took.

Move	Piles, biggest first
Start	10
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 10: _____

Move	Piles, biggest first
Start	5 5
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 5 5: _____

Move	Piles, biggest first
Start	2 2 2 2 2
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 2 2 2 2 2: _____

Problem 9: Run 3 counters from three piles of 1, 7 counters from one pile of 7, and 9 counters from piles of 5 and 4, each until a row repeats an earlier row. Then fill in the table for every number of counters from 3 to 10, using all your runs so far: does the game stop, or go round? If it goes round, how many arrangements are in each loop you found?

Move	Piles, biggest first
Start	1 1 1
1	
2	
3	
4	
5	

Move	Piles, biggest first
Start	7
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	5 4
1	
2	
3	
4	
5	
6	
7	
8	

Counters	Stops, or goes round?	Arrangements in each loop found
3		
4		
5		
6		
7		
8		
9		
10		

Problem 10: Find an arrangement of 10 counters that does not change at all when you make a move. Is there a second one? Then try to find an arrangement of 8 counters that does not change under a move. Explain why there is none: if a move changes nothing, which pile must be the new pile, and what does that force the other piles to be?

Problem 11: Look at your table in Problem 9: the game stopped for 3, 6 and 10 counters, at 2 1, 3 2 1 and 4 3 2 1. Write the next three numbers of counters for which you expect the game to stop, and the stairs it should stop at. Test the first of them: put that many counters in five piles of 3 and run until a row is the same as the row above it.

Next stopping numbers: _____, _____, _____ Stairs they stop at: _____

Move	Piles, biggest first
Start	3 3 3 3 3
1	
2	
3	
4	
5	
6	
7	
8	

Moves until it stopped: _____

Problem 12: Undo a move. Find every arrangement that turns into 3, 2, 1 after one move. Find every arrangement that turns into 5, 3, 2 after one move. Then try to find one that turns into 2, 2, 2, 2. If you cannot find one, write why no arrangement can turn into it.

Turn into 3 2 1: _____

Turn into 5 3 2: _____

Turns into 2 2 2 2: _____

Why not: _____

Problem 13: Sort these arrangements of 8 counters: which can be reached by a move from some arrangement, and which cannot? For each one that can, write an arrangement that reaches it. Then write what all the unreachable ones have in common, in a way that lets you decide for 6, 1, 1 and for 2, 2, 2, 1, 1, 1, 1 without running anything.

Arrangement	Reached from	Cannot be reached
4 4		
3 3 1 1		
2 2 2 2		
2 2 2 1 1		
5 1 1 1		
3 2 1 1 1		
3 1 1 1 1 1		
4 2 1 1		

6 1 1: _____ 2 2 2 1 1 1 1: _____

Problem 14: Find a start with 10 counters that takes more than 10 moves to reach the stairs. Write each start you try in a table and record its run. Continue on a blank sheet if you need more tries.

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Problem 15: The loop for 7 counters is 4 2 1, then 3 3 1, then 3 2 2, then 3 2 1 1, then back to 4 2 1. Build the stairs 3–2–1 with your counters. For each of the four loop arrangements, find where one extra counter of the second colour has to go to turn the stairs into it: on the tower of 3, on the tower of 2, on the tower of 1, or as a new pile of 1. Write it next to each. Then write all five arrangements that are the stairs 4–3–2–1 with one extra counter (11 counters). Start 11 counters at 5, 3, 2, 1 and make 6 moves. Does the game stay inside your list? Finally, write all six arrangements that are the stairs 3–2–1 with one extra counter on each of two different steps (a new pile of 1 counts as a step), and compare them with the two loops from Problems 4 and 5.

4 2 1: on _____ 3 3 1: on _____ 3 2 2: on _____ 3 2 1 1: _____

Stairs 4–3–2–1 plus one: _____

Move	Piles, biggest first
Start	5 3 2 1
1	
2	
3	
4	
5	
6	

Stairs 3–2–1 plus two: _____

Compared with the loops for 8 counters:

Problem 16: 12 counters. Write all ten arrangements that are the stairs 4–3–2–1 with one extra counter on each of two different steps. Predict how many loops there are for 12 counters: write the number in the blank, then write 1 next to each arrangement you predict is in the first loop, 2 next to each in the second loop, and so on. Then test: run 5, 4, 2, 1 for 6 moves and run 5, 3, 3, 1 for 6 moves. Are they in the same loop? Find a start outside your list that ends in the second loop.

Number of loops predicted: _____

Move	Piles, biggest first
Start	5 4 2 1
1	
2	
3	
4	
5	
6	

Move	Piles, biggest first
Start	5 3 3 1
1	
2	
3	
4	
5	
6	

Same loop? _____

A start outside the list that ends in the second loop:

Problem 17: How many moves does it take from 5, 5, 5 to reach the stairs 5–4–3–2–1? And from 4, 4, 3, 2, 1, 1? Record each run.

Move	Piles, biggest first
Start	5 5 5
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

Move	Piles, biggest first
Start	4 4 3 2 1 1
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	